The plug-in learner coefficient $\frac{1}{2}\operatorname{tr}(FH^{-1})$: sandwich covariance, excess loss and the information equality

Notes on Section 3.7 of the Stage 1 write-up

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The write-up predicts the regret slope of a plug-in (point-estimate) code by $\kappa = \frac{1}{2}\operatorname{tr}(FH^{-1})$, with H the Hessian of the expected loss and F the second moment of the per-example gradients at the optimum. These notes derive that formula in three steps (Sections 2–4), prove that $F = H$ for a correctly specified log-likelihood (Section 5), and list where each step can fail (Section 6).

1 Setup

Data $z_1, z_2, \ldots$ are i.i.d., $z = (x, y)$. The per-example loss is the negative log-likelihood of a conditional model,

$$\ell(z, \theta) = -\log p_\theta(y \mid x), \qquad \theta \in \mathbb{R}^d.$$

The population risk and its minimiser are

$$L(\theta) = \mathbb{E}\,\ell(z, \theta), \qquad \theta^* = \arg\min_\theta L(\theta), \qquad \nabla L(\theta^*) = 0,$$

and the empirical risk on m examples and its minimiser (the plug-in fit) are

$$L_m(\theta) = \frac{1}{m}\sum_{i=1}^{m} \ell(z_i, \theta), \qquad \hat\theta_m = \arg\min_\theta L_m(\theta), \qquad \nabla L_m(\hat\theta_m) = 0.$$

The two matrices, both evaluated at θ^*:

$$H = \nabla^2 L(\theta^*) = \mathbb{E}\,\nabla^2 \ell(z, \theta^*), \qquad F = \mathbb{E}\big[\nabla \ell(z, \theta^*)\, \nabla \ell(z, \theta^*)^\top\big].$$

H is the *curvature* of the expected loss: how fast it rises as θ leaves θ^*. F is the *spread* of the individual examples' gradients: because $\mathbb{E}\,\nabla \ell(z, \theta^*) = \nabla L(\theta^*) = 0$, it is their covariance, i.e. how much single examples disagree about which way to move. Both are computed from the same data distribution; they are different functionals of it, not the same quantity on different subsets.

2 The sandwich covariance

Lemma 1 (Covariance under a linear map)**.** *If g is a random vector with mean 0 and covariance Σ, and A is a fixed matrix, then* $\operatorname{Cov}(Ag) = A\Sigma A^\top$.

Proof. $\operatorname{Cov}(Ag) = \mathbb{E}[Ag\,(Ag)^\top] = A\,\mathbb{E}[gg^\top]\,A^\top = A\Sigma A^\top$. The matrix appears twice because a covariance is quadratic in the vector. □

Proposition 1 (Sandwich)**.** *Under the regularity conditions of Section 6 (in particular H positive definite),*

$$\sqrt{m}\,(\hat{\theta}_m - \theta^*) \xrightarrow{d} \mathcal{N}\big(0,\ H^{-1}FH^{-1}\big).$$

Derivation. Expand the first-order condition $\nabla L_m(\hat{\theta}_m) = 0$ around θ^* (mean-value form, $\bar{\theta}$ between $\hat{\theta}_m$ and θ^*):

$$0 = \nabla L_m(\theta^*) + \nabla^2 L_m(\bar{\theta})\,(\hat{\theta}_m - \theta^*) \quad \Longrightarrow \quad \hat{\theta}_m - \theta^* = -\big[\nabla^2 L_m(\bar{\theta})\big]^{-1}\,\nabla L_m(\theta^*).$$

Two limit theorems handle the two factors.

- *Law of large numbers.* $\hat{\theta}_m \to \theta^*$ (consistency), hence $\bar{\theta} \to \theta^*$, and the empirical Hessian converges to its mean: $\nabla^2 L_m(\bar{\theta}) \to H$.
- *Central limit theorem.* $\nabla L_m(\theta^*) = \frac{1}{m}\sum_i \nabla \ell(z_i, \theta^*)$ is an average of i.i.d. vectors with mean 0 and covariance F, so $\sqrt{m}\,\nabla L_m(\theta^*) \xrightarrow{d} \mathcal{N}(0, F)$.

By Slutsky's lemma, $\sqrt{m}(\hat{\theta}_m - \theta^*) \approx -H^{-1} \cdot \sqrt{m}\,\nabla L_m(\theta^*)$, and Lemma 1 with $A = -H^{-1}$ (symmetric) gives the covariance $H^{-1}FH^{-1}$. □

Remark (One dimension). With scalars $h = L''(\theta^*)$ and $f = \mathbb{E}[\ell'(z, \theta^*)^2]$, the derivation reads $\hat{\theta}_m - \theta^* \approx -g/h$ with g the average gradient, $\mathrm{Var}(g) = f/m$, so

$$\mathrm{Var}(\hat{\theta}_m) \approx \frac{f}{m\,h^2}.$$

Noisy per-example gradients (large f) make the fit wander; a sharply curved loss (large h) pins it down. The name "sandwich" only describes the shape $H^{-1}\,F\,H^{-1}$: F between two slices of H^{-1}.

3 The excess loss of the fit

The quantity a prequential code pays at step m is the expected loss of the fit $\hat{\theta}_m$ on a fresh example, in excess of the best predictor's:

$$\mathbb{E}\big[L(\hat{\theta}_m)\big] - L(\theta^*).$$

Expand L around θ^*; the linear term vanishes because $\nabla L(\theta^*) = 0$:

$$L(\hat{\theta}_m) - L(\theta^*) = \tfrac{1}{2}(\hat{\theta}_m - \theta^*)^\top H(\hat{\theta}_m - \theta^*) + o\big(\|\hat{\theta}_m - \theta^*\|^2\big).$$

For any random vector u with mean 0, $\mathbb{E}[u^\top H u] = \mathbb{E}\,\mathrm{tr}(Huu^\top) = \mathrm{tr}(H\,\mathrm{Cov}\,u)$. With $\mathrm{Cov}(\hat{\theta}_m - \theta^*) \approx H^{-1}FH^{-1}/m$ from Proposition 1:

$$\mathbb{E}\big[L(\hat{\theta}_m)\big] - L(\theta^*) \approx \frac{1}{2m}\,\mathrm{tr}\big(H\,H^{-1}FH^{-1}\big) = \frac{\mathrm{tr}(FH^{-1})}{2m} = \frac{\kappa}{m}, \qquad \boxed{\kappa = \tfrac{1}{2}\,\mathrm{tr}(FH^{-1})}.$$

In one dimension: $\frac{1}{2}h \cdot f/(mh^2) = f/(2mh)$. One factor of h cancels; what remains is the ratio of gradient noise to curvature.

Remark. Passing from the distributional limit to the expectation needs uniform integrability of $m\|\hat{\theta}_m - \theta^*\|^2$. This is the standard heuristic and holds in regular models; it is where heavy tails or rare divergent fits would break the formula.

4 From excess loss to the regret slope

A per-sample prequential code codes target $m+1$ with the fit on the first m targets. Summing the expected excess,

$$\mathbb{E}\,\mathrm{Reg}(n) \approx \sum_{m=1}^{n} \frac{\kappa}{m} = \kappa \ln n + O(1),$$

so κ is the slope of the regret against $\ln n$. (Block codes with geometric ratio r multiply this slope by $(r-1)/\ln r$; the write-up divides it out as the "batching correction".)

5 The information equality: $F = H$ under correct specification

Let $s(\theta) = \nabla_\theta \log p_\theta(y \mid x)$ be the score. Then

$$\nabla \ell = -s, \qquad \nabla^2 \ell = -\nabla \frac{\nabla p_\theta}{p_\theta} = -\frac{\nabla^2 p_\theta}{p_\theta} + \frac{\nabla p_\theta\, \nabla p_\theta^\top}{p_\theta^2} = -\frac{\nabla^2 p_\theta}{p_\theta} + s\, s^\top.$$

Taking expectations at θ^*:

$$H = F - \mathbb{E}\left[\frac{\nabla^2 p_{\theta^*}(y \mid x)}{p_{\theta^*}(y \mid x)}\right]. \tag{1}$$

Proposition 2 (Information equality)**.** *If the model is correctly specified at θ^*, i.e. the true conditional law of y given x is $p_{\theta^*}(\cdot \mid x)$ for (almost) every x, then $H = F$, both equal to the Fisher information $I(\theta^*) = \mathbb{E}[ss^\top]$.*

Proof. Condition on x and take the expectation over y under the true law, which is p_{θ^*} itself. The density cancels against the denominator in (1):

$$\mathbb{E}_{y \sim p_{\theta^*}(\cdot \mid x)}\left[\frac{\nabla^2 p_{\theta^*}(y \mid x)}{p_{\theta^*}(y \mid x)}\right] = \sum_y \nabla^2 p_\theta(y \mid x)\Big|_{\theta^*} = \nabla^2\Big(\sum_y p_\theta(y \mid x)\Big)\Big|_{\theta^*} = \nabla^2 1 = 0.$$

Averaging over x (under any distribution) gives $H = F$. □

Consequently $\mathrm{tr}(FH^{-1}) = \mathrm{tr}(I_d) = d$ and $\kappa = d/2$: the classical cost of $\frac{1}{2}\ln n$ per parameter, which in regular models coincides with the Bayes learning coefficient $\lambda = d/2$.

Assumptions of the information equality.

1. *Correct specification at θ^*.* The conditional law of y given x is exactly $p_{\theta^*}(\cdot \mid x)$. The distribution of x is arbitrary; only the y-expectation must be under the model's own law. This is the cancellation step.

2. *The loss is the log-likelihood.* No temperature, reweighting or penalty: for $\ell = -\tau \log p_\theta$, or with a regulariser added, $F \neq H$ even when the model is correct.

3. *Differentiation and summation commute,* and the support of p_θ does not depend on θ. Automatic for finitely many classes; for continuous y it needs dominated convergence.

4. *Smoothness and positivity.* p_θ is twice differentiable near θ^* and positive wherever the true law has mass (so that $\log p$ and s are defined).

5. *Only at θ^*.* Away from the optimum, or for the empirical versions at finite m, F and H differ; they agree only in expectation at θ^*, and the empirical estimates only asymptotically.

Example (Misspecified noise scale). Model $y \sim \mathcal{N}(\theta, \sigma^2)$, truth $y \sim \mathcal{N}(\mu, \tau^2)$. Then $\ell = (y - \theta)^2/(2\sigma^2) + \text{const}$, $\theta^* = \mu$, and

$$H = \frac{1}{\sigma^2}, \qquad F = \mathbb{E}\frac{(y-\mu)^2}{\sigma^4} = \frac{\tau^2}{\sigma^4}, \qquad \kappa = \frac{\tau^2}{2\sigma^2}.$$

The mean is estimated just as well either way, but an overconfident model ($\sigma < \tau$) pays more per $\ln n$ than $d/2 = \frac{1}{2}$, and an underconfident one less. Correct specification $\sigma = \tau$ restores $F = H$ and $\kappa = \frac{1}{2}$. Coding randomized-response labels with the clean softmax is the classification analogue of $\sigma < \tau$.

Remark (Empirical versus model Fisher). The write-up's F is the *empirical* Fisher: gradients evaluated at the observed labels. That is the F of the sandwich, the right quantity for κ. The *model* Fisher $\mathbb{E}_x\mathbb{E}_{y\sim p_{\theta^*}}[ss^\top]$ takes the y-expectation under the model; it equals F only under correct specification.

6 Where the steps fail

Assumptions of the sandwich and excess-loss steps (needed even when $F \neq H$): i.i.d. data; a unique optimum θ^* in the interior of the parameter set, to which $\hat{\theta}_m$ converges; ℓ twice continuously differentiable with dominated derivatives near θ^*; F finite; and H positive definite.

Singular models. If H is singular, H^{-1} does not exist: along the flat directions the fit does not concentrate at rate $m^{-1/2}$, the quadratic expansion is not the leading term, and Proposition 1 fails. This is the regime of Watanabe's theory, in which the Bayes learning coefficient satisfies $\lambda < d/2$ and a plug-in's coefficient need not equal λ. Exact gauge directions (such as $W \to WA$ for the subspace map) are harmless if they are quotiented out; the problem is directions that are flat to leading order but not exactly.

The write-up's networks. On GPT-2 the formula is applied with a pseudo-inverse on the non-flat directions. There $F \neq H$ for two reasons: the target law is only approximately realizable (it is reached only at infinite logits, so assumption 1 holds only in the limit), and the readout is saturated, so the per-example gradients are tiny in most directions. Few directions carry most of F, and $\frac{1}{2}\operatorname{tr}(FH^{-1})$ comes out two orders of magnitude below the family's nominal $\lambda = d_V(d - d_V)/2$: the code pays only for the directions the data actually pull on.